INCH-POUND
A-A-52447A
December 14, 2009
SUPERSEDING
A-A-52447
15 April 1993

COMMERCIAL ITEM DESCRIPTION

PUMP, FUEL, CAM-ACTUATED: ENGINE, MECHANICAL DIAPHRAGM AND PARTS KIT, ENGINE FUEL PUMP

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-P-45361B.

1. SCOPE

This commercial item description (CID) covers requirements for mechanical fuel pumps of the cam-actuated, diaphragm type used on carburetor-fed internal combustion engines. The rated flow of the fuel pumps ranges from 10 to 50 gallons per hour (gph). Requirements for repair kite to rebuild the fuel pumps are also covered.

2. SALIENT CHARACTERISTICS

2.1 Materials. Unless otherwise specified in the applicable engineering drawing (AED), the materials used shall be in accordance with the manufacturer's specifications. The use of recovered materials in compliance with regulatory requirements is acceptable provided that all the requirements of this CID are met (see 5.1, note a).

2.2 Design and construction. The design and-construction of the fuel pump shall be in accordance with the AED (see 5.1, note b). Requirements not specified in the AED shall be in accordance with the manufacturer's specifications/drawings.

Comments, suggestions, or questions on this document should be addressed to U.S. Army Tank-automotive and Armaments Command, ATTN: RDTA-EN/DM/STND/TRANS, MS# 268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.3 Repair parts kit. When specified (see 5.1, note c), the repair parts kit for the applicable fuel pump shall be comprised of the items listed either in the kit AED or in accordance with the manufacturer's kit specification. The repair kit, including instructions, shall be complete and adequate as required to rebuild and restore the fuel pump's performance as specified herein. The kit items shall not require hand fitting or machining for assembly into the fuel pump.

2.4 Performance. The fuel pump shall be capable of meeting all performance requirements when operated under conditions simulating intended use and installation on the vehicle. When specified (see 5.1, note d), the manufacturer's test procedures may be substituted for any of the verification procedures specified herein.

2.4.1 Shut off pressure. Unless otherwise specified in the AED, the fuel pump's shut off pressure shall be in accordance with the manufacturer's specification. The shut off pressure shall be verified by operating the fuel pump at any speed from 200 to 1800 cycles per minute (cpm) with no flow restriction for one minute. The flow shall then be restricted to zero for at least one minute and the shut off pressure verified at the fuel pump outlet.

2.4.2 Rated flow. The fuel pump's rated flow, in gph, at the indicated outlet pressures shall be as specified in table I. The rated flow shall be verified by operating the fuel pump at a speed of 1800 cpm for five minutes at each of the specified outlet pressures.

TABLE I. Flow rates.

Outlet pressure (percent of shut off pressure)	Percent rated flow
25	200
50	150
75	100

2.4.3 Endurance and vibration. The fuel pump shall be without failure for at least 500 hours while being subjected capable of operation to vibration resistance. This shall be verified while the fuel pump is mounted at an angle of 30 degrees to the right or left of the normal vertical operating position, as viewed in the cam shaft axial direction. The vibration produced shall be a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion). The frequency shall be varied uniformly from 10 to 55 Hertz (Hz) and back to 10 Hz in about 1 minute. The endurance/vibration cycle shall be as specified in table II.

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TABLE II. Endurance test factors.

Test Hours	Pump position			Rated flow (percent)	Outlet pressure (percent minimum shut off pressure)
	30 degrees right	Vertical	30 degrees left		
95		X		10	
1		X			75
95		X		20	
1		X			75
95		X		30	
1		X			75
95		X		40	
1		X			75
95		X		50	
1	X				75
6			X		75
6		X			75

2.4.4 Leakage. The fuel pump shall be capable of withstanding an internal air pressure of 10 pounds per square inch (psi) without any damage or leakage. This shall be verified by submerging the fuel section in water and the outlet port plugged.

2.4.5 High temperature. The fuel pump shall meet the performance requirements of d.1 through d.4 while operating in an ambient temperature of 160 degrees Fahrenheit (°F). This shall be verified by filling the fuel pump with a high flash point (140°F) mineral/petroleum type spirits solvent degreaser and conditioned for not less than 4 hours in an ambient temperature of 160°F prior to operation. The fuel pump's flow rate (see d.2) shall be met while operating in this ambient temperature except that the fluid test temperature shall not exceed 100°F.

2.4.6 Low temperature. The fuel pump shall meet the performance requirements of d.1 through d.4 while using a low temperature (below 32°F) gasoline and operating in an ambient temperature of minus 25°F. This shall be verified after the fuel pump and fuel have been conditioned for not 1ess than 16 hours in an ambient temperature of minus 65 ± 5°F. The fuel pump shall then be drained and operated. The fuel pump's delivery shall be verified as follows:

a. Fuel pumps with rated flow of 10 gph or less shall be capable of self-priming and delivering not less than two fluid ounces of fuel within the first 120 cycles of operation while operating at a speed of 200 cpm.

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b. Fuel pumps with rated flow of more than 10 gph shall be capable of self-priming and delivering not less than two fluid ounces of fuel within the first 40 cycles of operation while operating at a speed of 200 cpm.
c. The fuel pump shall be capable of meeting the performance requirements of salient characteristic d.1 while being operated at 200 cpm and under the conditions specified in a and b.

2.5 Identification and marking. Unless otherwise specified in the AED, identification and markings of fuel pump and parts kit shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code and part number and AED part numbers. Markings to designate the inlet and outlet ports shall also be provided.

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.1, note e).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this CID.
b. Title, number and date of the fuel pump AED and part number.
c. Whether repair parts kit is required and part number of the AED kit or manufacturer's kit.

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d. Whether manufacturer's test procedures and which ones are to be substituted for one or more of the performance verification procedures.
e. Selection of applicable level and packaging requirements.

5.2 Cross-reference data. Fuel pumps conforming to this CID are interchangeable/substitutable with fuel pumps conforming to MIL-P-45361B, dated 20 November 1985.

5.3 Metric product. The fuel pumps that are to metric dimensions will be considered on the following basis:

5.3.1 Unit conversion. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of IEEE SI10, and all other requirements of this CID are met.

5.3.2 Tolerance conversion. If a product is manufactured to metric dimensions and those dimension exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.3.2.1 Rejection. The contracting officer has the option of accepting or rejecting the product.

5.4 Regulatory requirements. Recovered material is material made in conformance to section 402 of the Clean Air Act (33 U.S.C. 1342 et. seq.).

5.5 Addresses for obtaining copies of referenced documents.

5.5.1 Government documents and publications. Copies of section 402 of the Clean Air Act (33 U.S.C. 1342 et. seq.) are available from Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402or website http://www.epa.gov/.

5.5.2 Non-Government publications. Copies of IEEE SI 10 International System of Units (SI): The Modern Metric System are available from the Institute of Electrical and Electronic Engineers, 445 Hoes Lane Piscataway, NJ, 08855-1331 or website http://www.ieee.org/.

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MILITARY INTERESTS:

Custodians
Army - AT
Air Force - 99

Review Activities:
Army – CR4
Air Force - 70
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

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